

Course Guide

Data Structures and Algorithms · هياكل البيانات والخوارزميات

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Every rule on this page comes from **the لائحة that governs your own program**, with the page number printed beside it. All four bylaws are in this repository under **regulations/**. Where this guide and your bylaw disagree, **the bylaw is right** — tell me and I will fix the guide.

1. Which course are you registered in?

Three programs, three bylaws, **one course**. Find your row.

Your program	Code	Bylaw	Spec page
Artificial Intelligence	CS2101	AI, 2020	p. 44
Bioinformatics / Medical Informatics	IS122	Medical Informatics, 2014	p. 35
Software Engineering	IS122	Software Engineering, 2013	p. 38

On the name "Bioinformatics". There is no Bioinformatics program at the faculty — the one usually meant is **Medical Informatics (المعلوماتية الطبية)**, and its bylaw is the 2014 file above. Bioinformatics is a course inside it (MED121, p. 45). If you were told you are in Bioinformatics, the 2014 bylaw is yours.

Side by side

	AI (2020)	Bio / Med. Inf. (2014)	SWE (2013)
Code	CS2101	IS122	IS122
Arabic name	هياكل البيانات و الخوارزميات	هياكل البيانات وتحليل الخوارزميات	هياكل البيانات وتحليل الخوارزميات
Credit hours	3	3	3
Weekly hours	2 lecture + 2 lab	2 lecture + 2 lab	2 lecture + 2 lab
Prerequisite	CS1002 Object Oriented Programming	CS012, MATH012	CS012, MATH012
Level / semester	Level 2, Semester 3	Level 2 (<i>semester not stated</i>)	Sophomore, Semester 1
Credits to graduate	138	135	135

Different codes. Same subject, same hours, same room, same exam paper.

2. What the لائحة says this course contains

The **2013** and **2014** texts are identical, word for word (p. 38 / p. 35):

*"Introduce the fundamental concepts of data structures and the algorithms that proceed from them. Topics include **recursion**, the underlying philosophy of **object-oriented programming**, fundamental data structures (including **stacks, queues, linked lists, hash tables, trees, and graphs**), the basics of **algorithmic analysis**, and an introduction to the **principles of language translation**."*

The **2020 AI** text (p. 44) is longer and names more structures:

*"This course provides fundamental data structures, algorithms, and **abstract data types** using mix of programming and theory. Main topics include data structures such as **arrays, lists, linked lists, stacks, queues, hash tables, heaps, priority queues, graphs, and trees**. It introduces algorithms such as those that are used for **list manipulation, graph searches, sorting, searching, and tree traversals**. Also, it introduces **analyzing and managing the complexity** associated with data structures and their operations."*

This course covers the union of both, so that every student in the room gets everything their own bylaw promises and a little more. [02-coverage.md](#) maps each declared topic to the module that implements it and the test that grades it.

Two things worth knowing about where this course sits

It assumes less than you might fear. The bylaw says this course *teaches* recursion and the philosophy of object orientation — it does not assume them. If you are in Bio, nothing before this course taught you either: CS012 is structured programming only, and MATH012 *Discrete*

Structures has no recursion, induction or asymptotics in it (2014, p. 30). That is exactly why Week 3 is recursion and why nothing here assumes you have written a class before.

For two of the three programs, this is the only algorithms course you will ever take. The AI bylaw has a follow-on — **AI3001 Analysis and Design of AI Algorithms** (2020, p. 48). The 2013 and 2014 bylaws have **none at all**. So if you are in SWE or Bio, what you do not learn here, you will not be taught anywhere else in your degree. Plan accordingly.

What this course unlocks

Course	AI	Bio	SWE
Database Systems	IS2102	IS123	IS123
Computer Networks	—	IT131	IT131
Computer Vision	—	IT132	IT132
Pattern Recognition	AI3301	IT137	—
Computer Graphics	AI2102	—	—
Software Construction	—	—	SWE141

SWE141 (2013, p. 42) is worth singling out: “BNF and basic theory of grammars and parsing... formal languages”. That is where this course’s “principles of language translation” grows up.

3. How you are assessed

All three bylaws agree on the three rules that matter.

Rule	Value
Final written exam	60% of the course mark
Coursework (أعمال فصلية)	the other 40%
To pass	≥ 60% overall and ≥ 30% of the final exam
Attendance	≥ 75% of lectures and labs, or محروم

Sources: SWE 2013 pp. 11–13, p. 18 · Med. Inf. 2014 pp. 11–13, p. 18 · AI 2020 pp. 14–16, p. 19.

The two rules that fail students

1 — The double threshold. You need **both**. A student who collects 35 of the 40 coursework marks and then scores 25% on the final has **failed**, however survivable the total looks. Coursework cannot rescue a final exam below 30%.

2 — Attendance. 75% of a 15-week course is roughly three sessions you can miss. Not five. Below that you are recorded as محروم and are not allowed into the final exam at all — it is not a penalty applied to your mark, it is a locked door.

How the 40% is split

The bylaws set bounds, not a fixed split:

- **SWE 2013 (p. 13) and Bio 2014 (p. 12):** midterm $\geq 20\%$
- **AI 2020 (p. 16):** midterm $\geq 10\%$, final $\geq 50\%$
- All three: no single component may **exceed 60%**

Not yet fixed. The exact split for this course will be announced in the lecture and on the [WhatsApp channel](#), and written here once it is. A split of **Final 60 · Midterm 20 · Coursework and practical 20** satisfies all three bylaws and is what I intend to propose. Until it is announced, treat only the four rules in the table above as binding.

Grade scale

Grade	Points	Percentage	Grade	Points	Percentage
A+	4.0	$\geq 97\%$	C+	2.3	$73 - < 76\%$
A	4.0	$93 - < 97\%$	C	2.0	$70 - < 73\%$
A–	3.7	$89 - < 93\%$	C–	1.7	$67 - < 70\%$
B+	3.3	$84 - < 89\%$	D+	1.3	$64 - < 67\%$
B	3.0	$80 - < 84\%$	D	1.0	$60 - < 64\%$
B–	2.7	$76 - < 80\%$	F	0.0	< 60%

Minimum pass in any course is **D**. Graduation needs a cumulative GPA of at least **2.00 / 4.00**, across **135** credit hours (SWE, Bio) or **138** (AI).

One conflict, documented rather than hidden: the 2013 bylaw values A+ at 4.33 on p. 14 and at 4.00 on p. 57. The other three documents all say 4.00, so 4.00 is used here. See [regulations/dsa-in-your-program.md](#).

4. What you need before week 2

```
git clone https://github.com/helghareeb/DSA27.git
cd DSA27
```

```
py -3.13 -m venv .venv
.\.venv\Scripts\activate
pip install -r requirements.txt
```

```
pytest -m "not challenge" # every one of these must pass
```

You also need **Graphviz** installed as a program — the dot binary, not only the pip package. The environment tests check for it.

Full setup notes, including the Windows “python opens the Microsoft Store” trap and the VS Code debugging setup, are in the repository [README.md](#).

5. How the course runs

Lecture (2 h/week)	The idea, its cost, and why it exists.
Lab (2 h/week)	You implement it. The skeletons in <code>dsa/</code> raise <code>NotImplementedError</code> .
Tests	<code>tests/</code> defines what “correct” means. <code>pytest -m challenge</code> .
Notebooks	One per week, in <code>notebooks/</code> , wired to the drawing helpers.
Visualisation	<code>viz/</code> draws, animates and measures. Already written — use it.

The rule that shapes everything, from `dsa/__init__.py`:

No module may use the built-in it is reimplementing.

`dsa/dynamic_array.py` may not be backed by a `list`. `dsa/hashmap.py` may not be backed by a `dict`. Routing around this may make tests pass; it will not make you a programmer.

The fifteen weeks are laid out in [01-study-plan.md](#).

Asking for help

1. Read the failing test. It states the contract.
2. Set a breakpoint and step through it — **F5**, *pytest: current test file*.
3. Then ask, on the [WhatsApp channel](#), with the failing test name and the error text.

“It does not work” is not a question. “`test_pop_from_empty_raises` expects `IndexError` and I am getting `None`” is.

6. From next year: the 2026 bylaw

The faculty has a new bylaw — [FCIS-Bylaw-2026.pdf](#), 328 pages, covering all seven programs. It replaces all three of the above with a single **CS201**: 3 credit hours, prerequisite CS101, Sophomore Fall, assessed 20/15/5/60, a major requirement in every program, and 142 credit hours to graduate (pp. 36, 138).

It does not govern you. It is in the repository so you can see where the faculty is going, and so this material is ready for the students who arrive under it.

7. Links

Repository	https://github.com/helghareeb/DSA27
WhatsApp channel	https://whatsapp.com/channel/0029Vb8XynEFy72KWbH6vS2V
YouTube	https://www.youtube.com/@0xHGH
Study plan	01-study-plan.md
Coverage matrix	02-coverage.md
Lecture 01	../lectures/01-why-this-course/lecture.md
Your program's regulations	regulations/
